

## Advanced (Habanero)

**Q1.** What is the vertex of the parabola  $y = x^2 + 4x + 4$ ?

- (A) (0, 4)
- (B) (0, 0)
- (C) (-2, 0)
- (D) (2, 0)
- (E) (-2, 4)

**Q2.** The axis of symmetry for  $y = x^2 - 6x + 9$  is  $x =$

- (A) -3
- (B) 0
- (C) 3
- (D) 6
- (E) 9

**Q3.** The parabola  $y = -x^2 + 2x - 1$  opens

- (A) up
- (B) down
- (C) left
- (D) right
- (E) vertically

**Q4.** What is the minimum value of  $y = x^2 - 4x + 7$ ?

- (A) -1
- (B) 0
- (C) 3
- (D) 4
- (E) 7

**Q5.** For  $y = 2x^2 - 4x - 6$ , the  $y$ -intercept is

- (A) (0, 2)
- (B) (0, -6)
- (C) (0, 6)
- (D) (2, 0)
- (E) (-2, 0)

**Q6.** What is the  $x$ -intercept of the parabola  $y = x^2 - 9$ ?

- (A) (0, 0)
- (B) (0, 3)
- (C) (3, 0)
- (D) (-3, 0)
- (E) (9, 0)

**Q7.** The axis of symmetry for  $y = 3x^2 + 12x + 15$  is  $x =$

- (A) -1
- (B) -2
- (C) -4
- (D) -6
- (E) -8

**Q8.** The vertex form of  $y = x^2 + 6x + 9$  is  $y =$

- (A)  $(x + 3)^2$
- (A)  $(x - 3)^2$
- (A)  $a(x - h)^2 + k$
- (A)  $(x + 6)^2$
- (A)  $x^2 + 6x$

**Open Ended Questions (FRQ)**

**Q9.** Sketch the graph of  $y = x^2 - 2x - 3$  and identify the vertex, axis of symmetry,  $x$ -intercepts, and  $y$ -intercept.

**Q10.** Find the maximum value of  $y = -x^2 + 4x - 3$  and the  $x$ -value at which this maximum occurs.

**Chef's Tips**

- Read each question carefully.
- Use a graphing calculator when necessary.
- Check your work.